Light, Pigment and Solvent

What is in the jar, and what it does to a photon.

Why this document exists. Its companion *Capture Fidelity* covers the **instrument** — how a webcam is made to yield a spectrum. This one covers everything *in front of* it: the light, the pigment, the solvent and the suspended matter. They are complementary halves, and for a long time only the first half was written down. That turned out to be a mistake: measured across a whole project, **the sample contributes more error than the camera does**.

Audience. A chemist who wants to know what the numbers mean physically; a developer who needs to know why the pipeline is shaped the way it is; and anyone reading a Spectracs report who asks what is actually being measured. No code appears anywhere in this document.

How to read it. Chapter 1 stands alone. Chapters 2 and 3 are the textbook part — light, matter and the pigments, none of it specific to us. Chapters 4 to 6 are where our particular sample gets awkward, and are the most practically useful. Chapter 7 says what follows for the measurement.

A note on numbers. Where a figure comes from our own measurements it is marked *(measured)* and the owning specification is named. Everything else is textbook physics or published chemistry, with sources in the appendix.

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1. The short version

1.1 What the instrument is really doing

A lamp shines through a jar of diluted pumpkin oil. A grating spreads the transmitted light into a spectrum, a camera photographs it, and the software divides that by the same measurement made without oil in the beam. The result is **absorbance** — how much light the sample removed, wavelength by wavelength.

Two features of the absorbance curve tell green oil from over-roasted oil: a strong band in the blue near 440–460 nm, and a weak one in the yellow-green near 560–580 nm. Their **ratio** is the verdict.

1.2 The five things worth remembering

> **1. Absorbance is a logarithm, and that is why it is useful.** Doubling the concentration doubles the absorbance, whereas it does *not* halve the transmitted light. Beer–Lambert linearity is what makes the numbers comparable at all (§2.2).
>
> **2. A ratio of two absorbance bands is almost indestructible.** Concentration, path length, lamp brightness and camera gain all scale absorbance *uniformly*, and a ratio divides a uniform scale straight out. This is why a €30 camera can do quantitative work (§2.3).
>
> **3. What survives a ratio is anything ADDITIVE.** Stray light, dark current — and above all **scattering by suspended matter**. These add to both bands and do not cancel (§2.5, §5).
>
> **4. The sample is not a solution.** A few drops of oil in isopropanol is a *dispersion*: the two are only partly miscible, so the oil arrives as suspended droplets alongside waxes that never dissolve at all. That suspension scatters, and the scattering masquerades as absorbance (§4, §5).
>
> **5. The pigment chemistry is simple; the sample preparation is not.** The pigment's two absorption bands and their fate on roasting are textbook — once you have the right molecule, which is **protochlorophyll**, not chlorophyll (§3.1). Everything difficult about this measurement lives in the jar, not in the molecule (§7).

1.3 The chain, in one paragraph

The pigment in the seed — **protochlorophyll**, chlorophyll's biosynthetic precursor — absorbs blue and red light and transmits green, which is why fresh Steirisches Kürbiskernöl is green. Roasting strips the magnesium out of the ring and the molecule becomes **protopheophytin**, which absorbs differently; more roasting degrades it further. So the shape of the blue absorption relative to the yellow-green absorption carries the roast history. Reading that shape requires diluting the oil, and diluting it into an alcohol produces a cloudy suspension whose light scattering sits underneath the pigment signal and compresses it. Most of the engineering effort in this project has gone into that last sentence.

2. Light and matter

2.1 What absorption actually is

A molecule absorbs a photon when the photon's energy matches the gap between two of the molecule's electronic energy levels. The energy of a photon is `E = hc/λ`, so **the wavelength that gets absorbed is set by the size of that gap** — a large gap absorbs blue light, a small gap absorbs red.

For the pigments that matter here the relevant electrons are those in **conjugated π systems**: alternating single and double bonds over which the electrons are delocalised. The longer the conjugated chain, the smaller the energy gap, and the redder the absorption. This one rule explains most of plant pigment colour, including why the carotenes are yellow-orange and the chlorophylls green.

An absorption *band* rather than a sharp line appears because each electronic transition is dressed with vibrational and rotational sub-levels, and in solution the surrounding molecules smear those further. In a liquid at room temperature the result is a smooth hump tens of nanometres wide — which is fortunate, because it means a modest instrument can resolve it.

2.2 Transmittance, absorbance and Beer–Lambert

Measure the light through the sample, `S`, and through the blank, `R`. The **transmittance** is their ratio, and the **absorbance** is its negative logarithm:

```
T = S / R                    A = −log₁₀(T) = log₁₀(R / S)
```

Absorbance is the useful quantity because of the **Beer–Lambert law**:

```
A = ε · c · l
```

— absorbance is *proportional* to concentration `c` and to path length `l`, with `ε` the molar absorption coefficient, a property of the substance at that wavelength. Transmittance is not proportional to anything: halving the concentration does not double `T`.

> **Where it breaks down.** Beer–Lambert assumes the absorbers are independent, dilute, and that only absorption removes light. At high concentration molecules interact and `ε` shifts; and any **stray light** reaching the detector puts a floor under `S`, so `A` stops rising with concentration and flattens. That flattening — *stray-light compression* — begins in practice below about `T = 10 %`, i.e. `A ≈ 1`. Our strongest band sits close to that line *(measured;* `SPEC_capture_quality.md` *§16.11.8)*, which is why the working dilution is chosen to keep it just under.

2.3 Why a ratio of two bands is so robust

If two bands are measured on the same sample, then `c` and `l` are identical for both, and

```
A₁ / A₂  =  (ε₁ · c · l) / (ε₂ · c · l)  =  ε₁ / ε₂
```

Concentration and path length cancel exactly. So does anything else that multiplies the whole curve — lamp brightness, camera gain, grating efficiency. What remains is a pure property of the substance.

This is why the pumpkin verdict is a *ratio* and not an absolute absorbance. It is also why the dilution recipe can be changed without recalibrating the verdict: measured across a 50 % change in concentration, the ratio moves by well under 1 % on green oil *(measured; `SPEC_capability_proof.md` §11.1)*.

2.4 What the reference cancels — and what it does not

Dividing by the blank is the single most important idea in the instrument. The lamp's own spectrum, the sensor's sensitivity at each wavelength, the grating's efficiency, the lens's vignetting — all of these appear **identically** in `R` and `S` and divide out exactly. That is why we do not need to characterise the camera.

But the cancellation is **multiplicative only**. Three things survive it:

survives the ratio	why	what we do
additive light — stray light, dark current	added after the sample, so it does not scale with `S`	dark measured and found negligible; stray light bounded by keeping `A ≲ 1`
non-linearity — the camera's gamma curve	a ratio of two *bent* numbers is not the ratio of the true ones	decode the curve before dividing
anything that changed between R and S — lamp drift, sensor warm-up, the jar moving	it was not the same instrument twice	measure R and S close in time, and don't disturb the jar

2.5 Scattering is not absorption — but the detector cannot tell

An absorbing molecule converts the photon's energy. A **scattering particle** merely sends the photon somewhere else. Physically these are completely different; optically, in a straight-through measurement, they are indistinguishable — **light that does not arrive is recorded as absorbance, whatever the reason.**

This matters far more than it sounds, and chapter 5 is devoted to it.

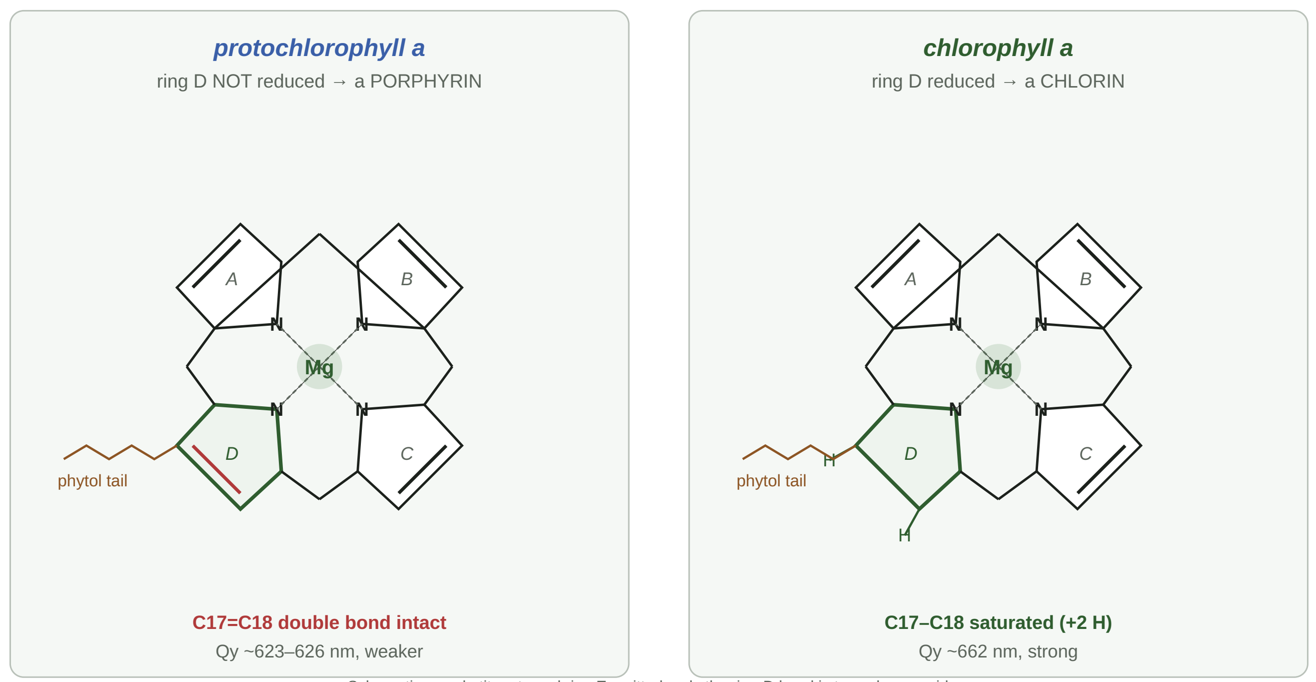
3. The pigments

3.1 The molecule in our jar is PROTOchlorophyll — not chlorophyll

This matters more than a prefix suggests, so it is worth being precise about.

Both molecules are tetrapyrroles: a large flat ring built from four nitrogen-containing subunits, with a magnesium ion held at the centre and a long hydrocarbon tail (phytol) that makes the whole thing fat-soluble. The difference is one bond.

	ring D	class	consequence
chlorophyll *a*	**reduced** (C17–C18 saturated)	a **chlorin**	strong, far-red Qy band
protochlorophyll *a*	**not reduced**	a **porphyrin**	Qy ~40 nm further to the blue, and weaker



Schematic — substituents and ring E omitted; only the ring-D bond is to scale as an idea.

Figure 1 — the macrocycle, and the single bond that separates the two molecules. Four pyrrole subunits (A–D) hold a magnesium ion between their four nitrogens; ring D carries the phytol ester that makes the pigment fat-soluble. In **protochlorophyll** the C17=C18 bond in ring D is intact, so the ring system is a **porphyrin**. In **chlorophyll** that one bond is reduced — two hydrogens added — making it a **chlorin**, and moving the red absorption band ~40 nm further into the red. Schematic: substituents and the isocyclic ring E are omitted.

Protochlorophyll is chlorophyll's biosynthetic *precursor* — the plant reduces that one ring to make chlorophyll. In the Styrian oil pumpkin's seed coat the reduction never happens, so the precursor accumulates. Fruhwirth & Hermetter identify the oil's colourants as **protochlorophyll (a and b)** and **protopheophytin (a and b)**.

3.2 Where the bands come from — Gouterman's four orbitals

All porphyrin-type spectra have the same two-part shape, and one model explains it. Gouterman (1961) showed that the visible spectrum is governed by just **four frontier orbitals** — two nearly-degenerate occupied and two nearly-degenerate empty ones. Their combinations give:

band	transition	character
Soret (also **B**)	$S_0 \rightarrow S_2$	**very strong**, blue, ~430–440 nm
Q	$S_0 \rightarrow S_1$	**weak** — 1/5 to 1/10 of the Soret — yellow-green to red

Blue and red are absorbed; the green in between is transmitted. That is the whole reason these pigments are green, and the reason fresh pumpkin seed oil is green.

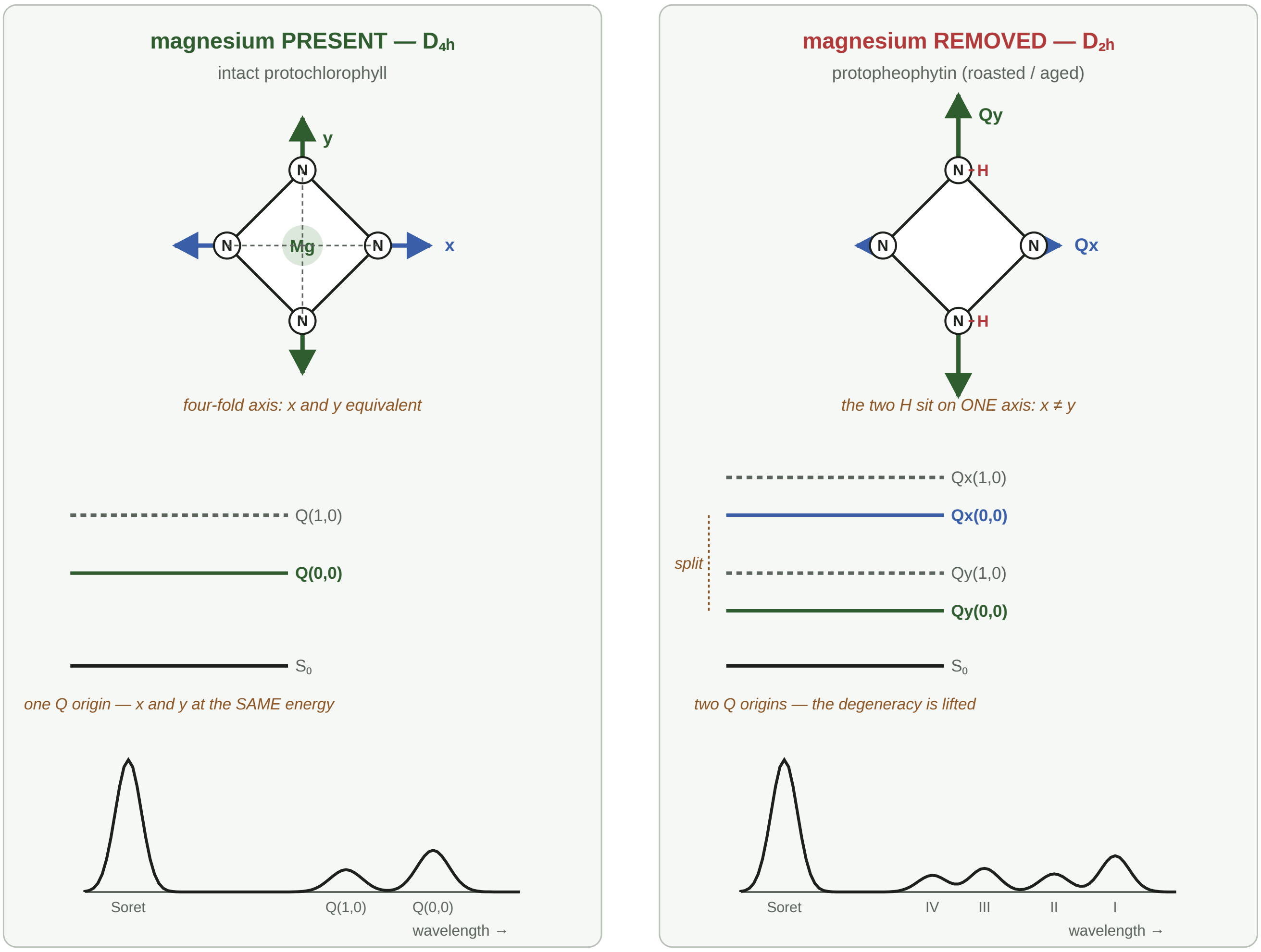
Why the Q region has more than one peak — and what Qx / Qy mean

In a **metallo**porphyrin the magnesium sits on a four-fold symmetry axis (D_{4h}). The two Q transitions are then **degenerate** — same energy, polarised at right angles to each other in the plane of the ring. What you see is one Q band plus a **vibronic satellite** at higher energy, conventionally labelled **Q(0,0)** (the origin, longest wavelength — also called α) and **Q(1,0)** (one quantum of vibration added — also called β).

Remove the metal and the symmetry drops to D_{2h}. The two protons that replace Mg^{2+} sit on one axis, so the ring is no longer four-fold symmetric, the degeneracy is **lifted**, and the two transitions separate into distinct bands — now properly called **Qx** and **Qy** after their polarisation axes. Each keeps its own vibronic satellite, so a metal-free (free-base) tetrapyrrole shows **four** Q bands, numbered **I to IV from the longest wavelength**: I = Qy(0,0), II = Qy(1,0), III = Qx(0,0), IV = Qx(1,0).

> **A word on "degenerate".** In spectroscopy *degenerate* means two states happen to share the same energy — it carries no sense of decay. Symmetry is what enforces the sharing, so lowering the symmetry **lifts** the degeneracy and the states separate. The pigment is not "degenerating"; its energy levels are ceasing to coincide.

⭐ **And the redistribution has a direction.** Free-base porphyrin spectra are classified into four types by their Q-band intensity ordering — *etio* (IV > III > II > I), *rhodo* (III > IV > II > I), *oxo-rhodo* and *phyllo*. **Band I — the longest-wavelength band — is the weakest in every one of them.** So a pigment whose long-wavelength Q band *dominates* while it holds its magnesium finds that same band demoted to the weakest of four once the magnesium goes. Intensity moves toward the blue. That is not a guess; it is the standard classification, and §3.4 shows it is what we measure.



Schematic. Free-base band numbering I–IV runs from the longest wavelength: I = Qy(0,0), II = Qy(1,0), III = Qx(0,0), IV = Qx(1,0).

Figure 2 — what losing the magnesium does. **Left:** with Mg on the four-fold axis (D_{4h}) the two in-plane transition dipoles *x* and *y* are equivalent, the Q states are degenerate, and the spectrum shows one Q origin plus its vibronic satellite. **Right:** the two protons that replace Mg sit on one axis only (D_{2h}), so *x* and *y* are no longer equivalent, the degeneracy is lifted into separate Qx and Qy states, and **four** Q bands appear where there were two. Free-base band numbering I–IV runs from the longest wavelength. Since demetallation is exactly what roasting does, this split is the spectroscopic signature of the degradation we measure.

> ⭐ **This is the single most useful idea in this chapter.** *Losing the magnesium is exactly what roasting does* (§3.3). So the **appearance of a Qx/Qy split is itself the spectroscopic signature of the degradation we are trying to measure.** The green→brown axis is not merely "less pigment" — it is a change in the *symmetry* of the surviving pigment, and symmetry changes rearrange bands rather than just shrinking them.

The numbers, for the molecule we actually have

	Soret	Q region	fluorescence
chlorophyll *a (textbook default — **not** ours)*	≈ 430 nm	≈ 578, **≈ 662 (Qy)**	≈ 668–675 nm
protochlorophyll(ide) *a* *(ours)*	**≈ 432–440 nm**	minor bands ≈ 505, 535, 606; **Qy(0,0) ≈ 623–626**	**≈ 630–636 nm**

The protochlorophyll figures are for alcoholic/acetone solution, which is what we measure in. They are corroborated from inside the oil itself: Fruhwirth & Hermetter record the oil's fluorescence maximum at **635**

nm, and a fluorescing molecule emits ~10 nm to the red of the transition it absorbs on — putting the absorber at ~625 nm and nowhere near chlorophyll's 662.

Where our three measurement windows sit

window	sits on	confidence
440–460 nm	the **red flank of the Soret band** — not its peak, which is below our usable range	solid
560–580 nm	a band in the **Q region**; the specific assignment is **open** — see below	⚠ open
600–630 nm	the approach to **Qy(0,0)** at ~623–626, plus the minor ~606 band	good

> ⚠ **An honest gap: we do not know what the 560–580 band is.** Two candidates, and they are not equivalent. It could be the **vibronic Q(1,0) satellite of the intact, Mg-containing protochlorophyll**; or it could be a **Qx band of protopheophytin**, the metal-free degradation product, which only exists *because* of the split described above. The published protochlorophyllide minor bands (505, 535, 606 nm) do not obviously include it, which mildly favours the second reading — but the oil contains both molecules plus carotenoids, and no source we have assigns this band.
>
> **Why it matters:** the 560–580 window is the *denominator* of the shipped verdict metric. If it is a degradation-product band, the metric is a direct intact ÷ degraded ratio and its physical justification is much stronger than we currently claim. **This is worth one measurement to settle** — and unusually, a hint already exists in our own data: relative to the 560–580 band, the 600–630 Qy flank is substantially *weaker* in brown oil than in green (ratio 0.58 vs 0.69), which is what you would expect if 600–630 tracks the intact pigment more specifically than 560–580 does. Suggestive, not conclusive.

3.3 What roasting does — protopheophytin

Heat and acid **strip the central magnesium ion out of the ring**, replacing it with two protons. The product is **protopheophytin**.

Where the name comes from. *Pheo-* is from the Greek *phaiós*, "dusky" — **pheophytin** is literally "the dusky plant pigment", the colour a chlorophyll turns once it loses its magnesium. It is the same reaction that makes overcooked greens go olive-drab. *Proto-* marks the unreduced ring D of §3.1. So the two prefixes describe **two independent modifications**, and there are four molecules:

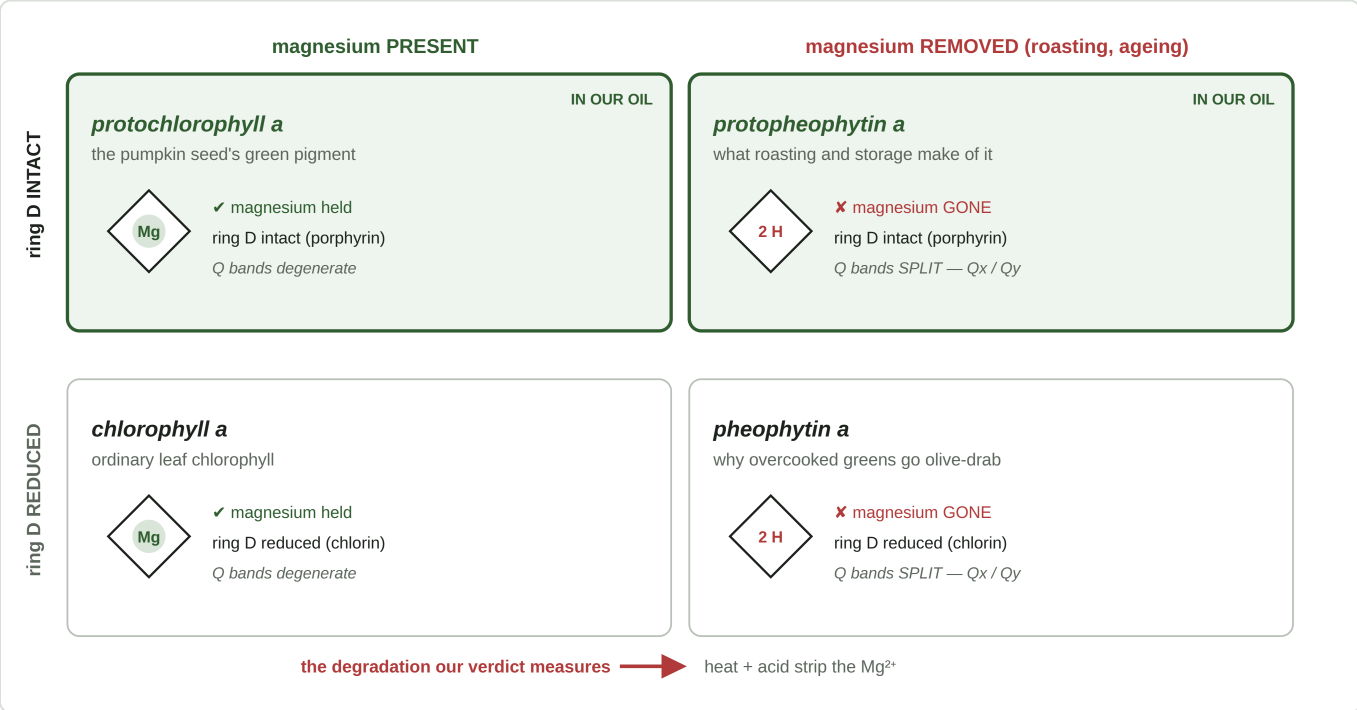


Figure 3 — the pigment family. Two independent changes, four molecules. Down the page: whether ring D is reduced (§3.1) — the difference between a porphyrin and a chlorin. Across the page: whether the magnesium is still held — the difference roasting makes. Our oil contains the **top row**: protochlorophyll and, as it degrades, protopheophytin. Ordinary leaf chlorophyll and its degradation product occupy the bottom row and are shown only for orientation.

Why protopheophytin matters here, and why it has moved to the centre of the story. It was named in the source all along — Fruhwirth & Hermetter list the oil's colourants as *"protochlorophyll (a and b) and protopheophytin (a and b), the latter being a protochlorophyll lacking the magnesium ion"*. What is new is the evidence that it is **spectroscopically active inside our measurement window**, which is §3.4.

It is not only roasting that makes it. In pumpkin seeds, protopheophytins accumulate as a **storage** degradation product and have been reported anywhere from **1 % to 36 %** of the protochlorophylls. That wide natural range is the chemical quantity the verdict is ultimately reading — and it is why the oil's history, not only its roast, shows up in the measurement.

Spectroscopically this is the $D_{4h} \rightarrow D_{2h}$ symmetry drop of §3.2: the Soret **weakens and shifts toward the blue**, and the Q region **gains structure** as the degeneracy lifts. Further degradation — oxidation, ring cleavage — eventually destroys the conjugated system altogether and the absorption decays to a featureless slope.

⭐ **This is the physical basis of the verdict, and it is worth stating in its strongest form.** We are **not** measuring "how green the oil is", nor even "how much pigment is left". We are measuring the **ratio of two chemical species** — intact protochlorophyll against its magnesium-free degradation product protopheophytin. Roasting and storage move that ratio; the ratio moves the spectrum; we read the spectrum.

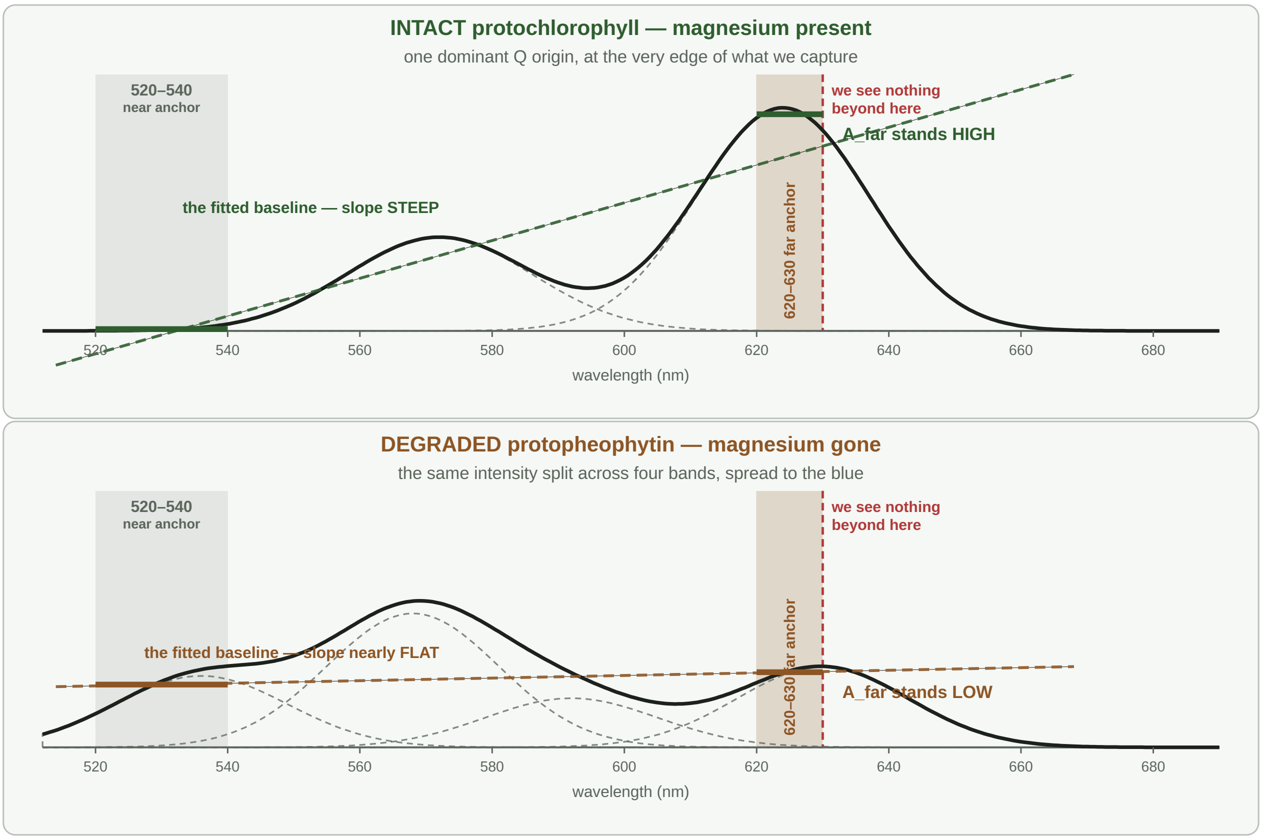
That is a stronger and more falsifiable claim than "less pigment", and the data insists on it: across our two classes the **total** Q-region absorbance is essentially identical (0.2300 vs 0.2251) while the *shape* differs at d = 10.3. Nothing is missing — something has been **converted**. It is a *roast/freshness* index, not a measure of "browning" in the Maillard sense.

3.4 Why this shows up as a change of SLOPE, not of level

Spectroscopically the $D_{4h} \rightarrow D_{2h}$ drop of §3.2 does something specific: it **redistributes** the Q intensity rather than removing it. One tall origin band becomes several smaller ones spread toward the blue. Total Q-region absorbance barely changes — which is exactly what we measure *(green 0.2300 vs brown 0.2251 over 560–580; `SPEC_capture_quality.md` §16.13)*.

So why does our 600–630 window see such a large difference? **Because that window is narrow and sits on a flank, not on a peak.**

Why redistributing the Q intensity LOWERS the 620–630 anchor — and FLATTENS the baseline through it



The far window sits on a FLANK, not on a peak, so its height reports the height of the nearest band — not the total pigment.

Measured: A(620–630) − A(520–540) = 0.0804 green vs 0.0488 brown (1.65×), over the 95.0 nm between the window centroids — 8.46 vs 5.14 × 10^{-4} A/nm, which is §5.3's fitted 8.56 / 5.22 to within 1 %. Schematic: pigment bands only, no turbidity pedestal, so its own contrast (17×) is far larger than the measured 1.65×.

⚠ The NEAR anchor is turbidity-dominated in reality, and the two classes nearly coincide there (0.1231 green vs 0.1038 brown), so the real pair of lines pivots about a common left foot. This model has no pedestal, so its left feet differ — read the RIGHT ends against each other.

Figure 4 — the mechanism. **Top:** with the magnesium intact, one dominant Q origin sits just past our capture limit, so the far red — and with it the **620–630 baseline anchor** — stands **high**, and the line fitted through the two anchor windows is **steep**. **Bottom:** once the magnesium is gone, that same intensity is spread across several weaker bands further to the blue; nothing tall is left near 630, the anchor drops, and the same line goes **nearly flat**. The total area under the two curves is similar; only its distribution differs. **The two solid bars are the only numbers the instrument takes from the curve** — one mean per anchor window — and the dashed line through them is the baseline it subtracts. ⚠ Schematic — the degraded band positions are illustrative, and with no turbidity pedestal modelled the contrast is exaggerated against the measured 1.65×; the two near anchors differ here where the real ones nearly coincide, so **compare the right-hand ends.**

The slope across a narrow window reports the height of the nearest peak, not the amount of pigment. That is the whole answer. A tall band whose edge crosses the window gives a steep rise; move that intensity 30–50 nm to the blue and split it, and the window is left on flat ground — even though just as much light is being absorbed overall, a little further to the blue.

This makes the far window an unusually specific probe: it responds to *whether the intact, symmetric pigment is still there*, and is comparatively blind to how much total pigment the oil contains.

⚠ **Measured, and it is not a concentration effect.** Under simple Beer–Lambert the two classes' curves would differ only by a scale factor, so any ratio taken *inside* the Q region would be identical for both.

Measured on the two post-rebuild sets, the 620–630 rise divided by the Q band's own amplitude is **0.427 for green against 0.080 for brown — a factor of 5.3, at *d* = 10.3**. Scaling is excluded; the shape genuinely differs. *(`SPEC_capture_quality.md` §16.13.)*

⚠ Why we do not simply *look* at the two-versus-four bands — and it is not the instrument

The natural objection is that we are inferring a band structure we cannot see, and that a better spectrometer would settle it. **It would not.** Measured on our own data:

grid spacing	0.146 nm per bin, 1305 bins across 440–630 nm
narrowest feature we resolve *(473 nm lamp artifact)*	**FWHM 1.0 nm**
second artifact *(607 nm registration)*	**FWHM 2.7 nm**
the Q bands we would need to separate	**20–30 nm wide**

The rig **out-resolves the target by ten to twenty times**. Four other things stand in the way instead:

1. **Window truncation — the dominant limit.** Our capture stops at 630 nm and the longest-wavelength band sits at ~623–630. We see a flank and nothing beyond it.
2. **The two species always coexist.** Real oil is a mixture (protopheophytin 1–36 %), so a pure two-band or pure four-band spectrum is never presented to us — only their superposition.
3. **Intrinsic linewidth.** At room temperature in solution these bands are 20–30 nm wide and overlap heavily. Even with a perfect instrument and an unlimited window, the four free-base bands appear as partly merged shoulders, not four clean peaks.
4. **The turbidity pedestal**, which contributes 52–61 % of the Q band's height and flattens what contrast remains.

⇒ The remedy is a wider window, not a finer one. That single conclusion redirects an entire class of "buy better hardware" thinking toward a much cheaper change.

⚠ What is measured, and what is inferred

link	status
the pigments are protochlorophyll and protopheophytin	**sourced**
roasting and storage strip the magnesium	**sourced**
losing the metal lifts the degeneracy: two Q bands become four	**deduction** from symmetry
free-base band I is the weakest ⇒ long-wavelength intensity falls	**sourced** (the four-type classification)
the two classes' Q-region shapes genuinely differ	**measured**, *d* = 10.3
the cause is *specifically* demetallation	⚠ **best available explanation, not a controlled result**

The last row is the honest weak point: these are two different **bottles**, which differ in more than their protopheophytin content. Note what survives if it is wrong — the *speciation* reading holds regardless; only the *mechanism* would need replacing.

The experiment that would close it is cheap. Take one oil, split it, and deliberately demetallate half — acidification is the standard laboratory route. Same bottle, same turbidity, same dilution, one variable. If the

600–630 slope collapses in the acidified half, the causal link stops being an interpretation and becomes a measurement.

> ⭐ **Read the slope on BOTH windows when that run happens.** Since 2026-08-03 the shipped far anchor is **620–630 nm**, not 600–630 (`SPEC_capture_quality.md` §16.20). The prediction above is about the *pigment*, so it is anchor-agnostic and stands as written — but 620–630 is what the instrument now acts on, and it is only 10 nm wide, so it is also the noisier of the two. Recording both costs nothing (the diagnostics compute both) and keeps the prediction falsifiable on the window that matters.

3.5 The carotenoids

The second pigment family is the **carotenoids** — β-carotene, lutein and relatives. These are long conjugated polyenes with no ring metal, and they absorb in a broad three-peaked structure across **400–500 nm**, with essentially nothing above about 520 nm. They are what makes carrot juice orange and pumpkin flesh deep yellow.

For us they are mostly a nuisance: their absorption tail reaches into the region we would like to use as a quiet baseline, and unlike the suspended matter of chapter 5, **it is real absorbance that no amount of clarification will remove**.

3.6 Why pumpkin seed oil is green — or brown

Styrian pumpkin seed oil contains chlorophyll and its derivatives from the seed coat, together with carotenoids. A fresh, gently pressed oil retains enough intact chlorophyll to look distinctly green in a thin layer. Longer or hotter roasting pheophytinises it, and the green retreats.

The commercial question — *"was this oil over-roasted?"* — is therefore a question about a molecular ratio, and that is a question a spectrometer is genuinely better at than an eye.

3.7 Dichromatism: green in a film, red in the bottle

Hold Kürbiskernöl in a thin film and it is green. Look through the bottle and it is deep red. Nothing about the oil changed — **the path length did.**

This is **dichromatism**, and pumpkin seed oil is its textbook example. The oil has a narrow transmission window in the green plus a broad one in the red. In a thin layer both get through and green dominates because the eye is most sensitive there; as the path lengthens, the narrow green window is extinguished exponentially faster than the wide red one, and red wins. Kreft and Kreft quantified the effect as a **dichromaticity index**.

> **Why it matters practically.** Perceived colour depends on path length, so *"it looks green"* is not a measurement. It also means our transmission-derived colour swatches are dilution-dependent by construction, while absorbance-derived ones are not — a distinction the evaluation code takes care to keep *(`SPEC_color_retrieval.md`)*.

4. The sample: oil in a solvent

4.1 Why dilute at all

Neat pumpkin oil is effectively opaque across the pigment bands at any practical path length. To bring the absorbance into the instrument's usable range — roughly `A = 0.1` to `1.0` — the oil must be diluted perhaps thirty-fold. **The solvent is therefore part of the measurement, not a convenience**, and the choice of solvent has consequences all the way to the verdict.

4.2 Oil and alcohol are only partly miscible

Pumpkin oil is a **triacylglycerol**: three long fatty-acid chains on a glycerol backbone, overwhelmingly nonpolar. Isopropanol is *semi-polar* — a polar hydroxyl group on a small nonpolar isopropyl group. "Like dissolves like" is a crude rule but here it bites: the two are only **partially miscible**.

Such a pair has an **upper critical solution temperature (UCST)**. Above it they mix in all proportions; below it there is a **miscibility gap** — a range of compositions where a single phase is not stable and the mixture separates. Solubility rises with temperature up to the critical point.

> **Water is the hidden variable.** Adding water to the alcohol raises the critical temperature sharply, i.e. makes the solvent markedly worse. Isopropanol is **hygroscopic**, so an opened bottle takes up atmospheric water over months. A tired bottle of "99 %" is measurably worse at dissolving oil than a fresh one — and nothing on the label says so.

4.3 The ouzo effect

When a solution of oil in a *good* solvent is rapidly diluted with a *poor* one, the oil finds itself suddenly far above its solubility and comes out of solution — not as a separated layer, but as a cloud of **spontaneously formed droplets**, typically tens to a few hundred nanometres across. The mixture turns milky within seconds without any stirring energy being supplied.

This is the **ouzo effect**, named for what happens when water is added to anise spirits. It occurs in the metastable region between the binodal and spinodal curves of the phase diagram, and it needs no surfactant.

Adding a few drops of oil to isopropanol is the same operation in miniature: the oil is locally concentrated at the point of entry and is then diluted below its solubility. **What forms is not a solution but a metastable dispersion.**

4.4 What happens next — ripening, coalescence, sedimentation

A fresh dispersion is not stable, and it does not simply sit there. Three processes run in parallel:

- **Ostwald ripening.** Small droplets have higher internal pressure and slightly higher solubility than large ones, so material diffuses from small to large. The droplets grow, and the population coarsens.
- **Coalescence.** With no surfactant to keep them apart, droplets that meet merge.
- **Sedimentation.** Once droplets are large enough, gravity wins over Brownian motion.

And here they sink rather than float. Oil is denser than isopropanol — about 0.92 against 0.785 g/mL — so the droplets fall. (In water the same droplets would cream upward; the direction is set by the density difference, not by the oil.)

The rate follows **Stokes' law**:

```
v = (2/9) · Δρ · g · r² / η
```

The `r²` is the important part. Fine droplets settle imperceptibly; as ripening grows them, the settling accelerates **non-linearly**. That is why a fresh dilution appears stable for a while and then clears faster and faster — and why "how long has it stood?" is a question with a strongly non-linear answer.

> **Observed here.** A fresh dilution's readings drift for the first ~15 minutes and then continue changing for hours, **non-monotonically**: our measured ratio falls over the first half hour and rises over eleven *(measured; `SPEC_capability_proof.md` §11.4a–f)*. Stirring an aged sample re-suspends the sediment and puts the reading straight back where it started.

4.5 Choosing a solvent: polarity and solvency are different axes

The useful and slightly counter-intuitive point: **how well a solvent dissolves oil is not the same as how polar it is.**

- **Solvency** for a triglyceride tracks the **alkyl chain length** — a longer hydrocarbon tail on the solvent resembles the fatty-acid chains it has to surround.
- **Solvatochromic shifts** — the small movements of an absorption band caused by the solvent — track the **dielectric constant** `ε`.

These can be varied nearly independently, and that is a lever.

solvent	ε (25 °C)	b.p. °C	flash pt °C	dissolves oil	polystyrene	usable here
water	78	100	—	not at all	✅	—
2-propanol *(in use, and staying)*	**≈ 17.9**	82.6	12	**marginal**	✅ safe	✅ **chosen**
1-propanol	≈ 20.1	97.2	22	better	✅ safe	⛔ H318
1-butanol	≈ 17.5	117.7	35	**good**	✅ safe	⛔ **H318**
2-butanol	≈ 16	99.5	24	between the two above	~ caution	⛔ not pursued
acetone	≈ 20.7	56.1	**−20**	good	⛔ **dissolves it**	⛔ not pursued
n-heptane	1.9	98.4	**−4**	ideal	⛔ swells + crazes	⛔ hazardous
cyclohexane, isooctane	≈ 2	—	—	ideal	⛔ cyclohexane **dissolves** it	⛔

In principle a longer-chain alcohol is the interesting direction. Its dielectric constant is close to isopropanol's, so absorption bands should barely move, while its longer chain makes it a better triglyceride solvent — dissolution without solvatochromism.

4.6 ✅ The decision: isopropanol stays

The instrument keeps 2-propanol and its existing polystyrene vessel. Not because the alternatives were untested in principle, but because each fails on a specific, checkable ground — and because the problem the swap would solve stopped being the one that limits the instrument.

candidate	why it is not used
1-butanol *(and 1-propanol, isobutanol)*	**H318 — "causes serious eye damage", Category 1: irreversible.** For an instrument meant to be operated by a miller in food premises, that is the wrong classification to accept. *(tert-butanol is excluded separately: it melts at 25.8 °C and is a solid at room temperature.)*
2-butanol	The one butanol both liquid at room temperature and free of Cat-1 eye damage — but a **branched** alcohol, so its solvency gain over isopropanol is only partial; more volatile; and its ε near 16 rather than 17.8 weakens the very "bands barely move" argument that made a butanol attractive. Not enough gain for the disruption.
acetone	Chemically appealing — 80 % acetone is the *standard* solvent in the pigment literature (§3.2), so it would place our band positions directly on the published scale. But it **dissolves polystyrene**, so it demands a new vessel, and no suitable one exists (§6.6).
n-heptane, cyclohexane	Ideal solvents, but they attack the vessel and carry hazards no food producer should be asked to keep on a shelf.

And the reason the question is closed rather than merely deferred: the solvent work existed to reduce the scattering pedestal — that is, to buy **precision**. Precision is no longer what limits this instrument. The measured separation between a green and a brown oil is roughly **eleven standard deviations**, with no overlap between any two runs *(`SPEC_capture_quality.md` §16.13)*. What limits the instrument now is whether the **threshold** dividing the two classes is in the right place — a question no solvent can answer, and which a solvent change would actively set back, since every band position and therefore the threshold itself would have to be re-derived.

> ⚠ **One condition attaches.** All of that rests on the *sample-preparation* scatter being modest, and that quantity has not yet been measured (§5.5's aliquot trap). If it turns out to be large, turbidity becomes the limiting term again and the solvent question genuinely reopens.

4.7 What the analytical literature actually uses

Standard methods for chlorophyll and carotenoids in edible oils read them in **cyclohexane**, **hexane**, **carbon tetrachloride**, or **ethanol/isooctane** and **ethanol/heptane** mixtures. Nobody uses neat isopropanol.

The field converged on hydrocarbons precisely because they give a true solution — no dispersion, no scattering, no settling. Our choice of an alcohol is a deliberate trade for field usability, and chapter 5 is the bill for it.

> **On waiting.** Two pieces of standard guidance are often quoted and neither applies to us. *"Measure turbid samples within ten minutes"* comes from **nephelometry**, where the particles *are* the analyte and settling loses the signal. *"Allow 10–15 minutes to equilibrate"* refers to thermal equilibration and colour development in wet-chemistry assays. Where particles are the **interferent**, as here, the standard practice is neither — it is **to clarify: filter or centrifuge before measuring**, then fit a baseline for whatever residual remains.

5. Turbidity: the pedestal

5.1 Scattered light is recorded as absorbance

Restating §2.5 because everything in this chapter follows from it: a straight-through spectrometer measures **light that failed to arrive**. It cannot ask why. Light removed by scattering from a suspended droplet is therefore added to the absorbance, exactly as though a pigment had absorbed it.

Because the scattering is broad and smooth in wavelength, it lifts the **whole** absorbance curve off zero. We call that additive floor the **pedestal**.

5.2 Particle size sets the colour of the scatter

How the scattering depends on wavelength is governed by the particle size relative to the wavelength:

regime	particle size	wavelength dependence	appearance
Rayleigh	much smaller than λ	**$\propto \lambda^{-4}$** — strongly blue-biased	the blue sky; a faint blue haze
Mie	comparable to λ	weakly dependent, oscillatory	milky white
geometric	much larger than λ	essentially flat	white, opaque

So the *shape* of the pedestal is a fingerprint of what is suspended: a λ^{-4} tilt points to nanodroplets, a flat floor to micron-scale waxes and press fines. In principle this is measurable; in our spectrum it is frustrated by the absence of a genuinely pigment-free window to fit through *(`SPEC_capture_quality.md` §16.12.11)*.

5.3 How large it is here, and why it wrecks a ratio

Measured across eight independent sample preparations, the pedestal in our weaker measurement band runs at **0.7 to 1.9 times the pigment signal itself** *(measured; `SPEC_capability_proof.md` §11.4e)*. **The plinth is bigger than the statue standing on it.**

And it is specifically corrosive to a *ratio*, because adding the same amount `c` to numerator and denominator drags any ratio toward 1:

```
true ratio        12.37 / 1.00                       = 12.37
with pedestal    (12.37 + 1.59) / (1.00 + 1.59)     =  5.39
```

The pigment did not change. The pedestal compressed the reading by more than half. **This is the entire difference between the two pigment-ratio figures a Spectracs report prints** — one before the baseline correction, one after — and it is why a baseline correction exists at all.

Note what this does to §2.3's reassuring algebra: a ratio is immune to everything *multiplicative*, and turbidity is the classic *additive* offender. Robustness has a specific shape, and this is the hole in it.

5.4 An accidental experiment: three years on a shelf

The clearest demonstration was not designed. Oils bought in 2023 and measured in 2026 had spent three years in the bottle — during which the same sedimentation described in §4.4 quietly clarified them. Fresh 2026 oils had not.

	pedestal	run-to-run scatter
oils aged three years	**0.84**	2.5 %
fresh oils	**1.72**	14.5 %

The aged oils carried **half** the scatter — and separated the two quality classes about **eight times better** *(measured;* `SPEC_capability_proof.md` *§11.4e)*.

> **The lesson, and it is the most useful sentence in this document: sample clarity is a first-class instrument parameter.** It is not housekeeping. On the evidence above it is worth more than any optical or algorithmic improvement available to us.

5.5 Getting rid of it

Four routes, in rough order of attractiveness:

1. **A better solvent.** If the oil truly dissolves, there is nothing to scatter. This is why §4.5's distinction between polarity and solvency matters — but **this route is closed for this instrument** (§4.6, §6.6): every solvent that dissolves the oil properly either carries an unacceptable hazard classification or attacks the vessel, and the vessel sits above a mains-voltage lamp.
2. **Filtration.** A syringe filter removes anything above its pore size. Note the limit: ouzo nanodroplets at 50–200 nm pass straight through a 0.22 µm membrane, so a filter distinguishes *which* population is responsible as much as it removes it.
3. **Centrifugation.** Faster and consumable-free, at the cost of another instrument and a transfer step.
4. **Waiting.** Free, slow, non-monotonic, and it leaves the sediment in the vessel where any agitation will re-suspend it. The weakest of the four, and where we started.

> **A trap worth naming.** If the sample is mixed in one vessel and an aliquot is transferred to another for measurement, **the transfer is itself a sampling step out of a settling dispersion**. How much suspended matter travels with the aliquot depends on when and from what depth it was drawn. Homogenise before drawing, or clarify after; doing neither leaves a large, invisible sample-to-sample variable *(*`SPEC_capability_proof.md` *§11.4f)*.

6. The vessel

6.1 Plastics and solvents

The sample sits in a small transparent jar. Which materials survive contact is decided by the same "like dissolves like" rule as §4.2, now working against us:

material	alcohols	aliphatic hydrocarbons	note
polystyrene	✅ resistant	⛔ **attacked**	cheap, optically clear, what we use
PET / PETG	✅	mostly ✅	good general compromise
polypropylene	✅	fair, swells slowly	translucent, poor optically
PTFE / FEP	✅	✅	inert to essentially everything
borosilicate glass	✅	✅	inert, but see §6.5

Alcohols are essentially the only organic family polystyrene tolerates. That couples the solvent and the vessel into a single decision: stay with alcohols and the cheap clear jar keeps working; move to a hydrocarbon and the vessel must be replaced too.

6.2 Putting a number on it: Hansen parameters and RED

Compatibility charts like the one above are qualitative, and at the margins they are unreliable — published charts contradict one another on cases such as acrylic against aliphatics. **Hansen solubility parameter theory replaces the judgement with arithmetic.**

The idea is that "like dissolves like" can be made three-dimensional. Every substance gets three coordinates: `δD` for dispersion forces, `δP` for polar interactions, `δH` for hydrogen bonding. A *polymer* is not a point but a **sphere** of radius `R₀` — the region of that space whose solvents dissolve it. The distance from a solvent to the polymer's centre is

```
Ra² = 4·(δD₁ − δD₂)² + (δP₁ − δP₂)² + (δH₁ − δH₂)²
```

(the factor 4 on the dispersion term is empirical — it is what makes the solubility region come out spherical rather than ellipsoidal), and the useful quantity is that distance scaled by the sphere:

```
RED = Ra / R₀              "Relative Energy Difference"
```

RED	meaning
< 1	inside the sphere — the solvent dissolves the polymer
≈ 1	on the boundary — swelling, crazing, environmental stress cracking
> 1	outside — a non-solvent

Against **polystyrene** (δD 21.3, δP 5.8, δH 4.3, R_0 12.7):

solvent	RED	reading
toluene	**0.65**	dissolves it

solvent	RED	reading
cyclohexane	**0.90**	**dissolves it** — the classic theta-solvent for polystyrene
n-heptane	**1.10**	just outside — swells and stress-cracks, does **not** dissolve
1-butanol	**1.23**	outside — safe; the closest of the alcohols
2-butanol	⚠ **not computed**	likely **closer to PS** than 1-butanol — see below; not pursued (§4.6)
isooctane	1.27	outside
2-propanol	**1.29**	outside — safe
1-propanol	1.33	outside
ethanol	1.49	comfortably outside
water	3.23	inert

> ⚠ **2-butanol's RED is deliberately left blank rather than estimated.** Computing it needs the same source's polystyrene parameters, and a value derived from a different source would not be comparable with the rows above. **But the direction is predictable and it is the wrong one:** 2-butanol is a *secondary* alcohol, so its hydrogen-bonding term δH is **lower** than 1-butanol's — which moves it **toward** polystyrene in Hansen space, i.e. toward the swelling boundary that heptane sits on at 1.10. A resin-compatibility guide independently rates 2-butanol against PS only as *"moderate–good, with caution"*, against 1-butanol's clean "safe at 20 °C". **Two independent hints in the same direction ⇒ the overnight soak test (`SPEC_capture_quality.md` §16.12.7 item b) is a gate, not a precaution.**

Three things worth taking from that table:

- **The alcohols are not borderline.** Butanol at 1.23 against isopropanol's 1.29 is a small difference well outside the sphere, not a near-miss.
- **Aliphatic hydrocarbons do not dissolve polystyrene** — heptane at 1.10 swells and crazes it. The practical verdict is the same, but the mechanism is not, and confusing the two leads to the wrong test.
- **Cyclohexane, which the analytical literature favours for oil pigments, is the worst possible choice for a polystyrene vessel** — at 0.90 it is a genuine solvent for it. A reminder that the best solvent for the *sample* and the best for the *container* are separate questions.

> ⚠ **Do not mix parameter sets.** Published values differ by a few percent, and polystyrene's `δD` is quoted as either ≈ 18.6 or ≈ 21.3 depending on convention. Combining a solvent from one set with a polymer from another produces confidently wrong answers. The *ranking* is robust; treat absolute values as indicative and always compare within a single source.

6.3 Crazing, and why the failure mode is optical

The realistic failure is not a dissolved jar. It is **environmental stress cracking**: a marginal liquid plus the frozen-in stress of an injection-moulded part, producing a network of fine surface crazes. It is cumulative in exposure time, and it starts where the stress is highest — around threads and gate marks rather than in the middle of a wall.

A crazed vessel does not crack. It goes very slightly hazy. And haze scatters light, which this instrument records as absorbance (§5.1).

> **This is the nastiest failure mode in the whole document**, because a slowly crazing jar would look exactly like sample turbidity — the very thing one changes solvent to remove. The experiment and its control would degrade together, and the conclusion drawn would be that the new solvent had not worked.
>
> The check is cheap and uses the instrument itself: cycle a spare vessel through twenty fill-and-empty cycles at realistic contact time, then **measure it as a blank against an unexposed one**. A raised baseline is crazing, detected far below the threshold of the eye.

6.4 Refractive index, reflections and the meniscus

Every interface between two materials of different refractive index reflects a little light. For light arriving straight on, the reflected fraction is

```
R = ((n₁ − n₂) / (n₁ + n₂))²
```

interface	$n_1 \rightarrow n_2$	reflected
liquid → **air**	1.377 → 1.00	**2.5 %**
liquid → polystyrene	1.377 → 1.59	0.5 %
liquid → borosilicate	1.377 → 1.47	0.1 %
liquid → **FEP film**	1.377 → 1.344	**0.015 %**

Two things follow. First, **a closely index-matched window is almost invisible optically** — fluoropolymer film against an alcohol reflects a hundred and fifty times less than a liquid–air surface. Second, and more importantly, a liquid–air surface is a **meniscus**: a curved surface whose shape changes with fill level, wetting and tilt. It is a lens that is slightly different every time. Eliminating it — by letting a window contact the liquid — removes a variable, not just a reflection.

6.5 The awkward geometric constraint

Because the light passes through the vessel *and* its lid, **both must be transparent** — and a clear vessel with a matching clear lid is not something one simply buys in glass. The usual answer would be a lid with an aperture carrying a clamped **fluorinated ethylene propylene (FEP)** film: better than 95 % transmission across the visible, under 2 % haze, chemically immune to everything discussed here, and clamped rather than glued, since the non-stick nature that makes it solvent-proof also means nothing adheres to it.

6.6 ✅ The decision: the polystyrene vessel stays — and a safety constraint decides it

The instrument keeps its existing sealed polystyrene jar. Two reasons, and the second is the one that settles it.

First, nothing better is available off the shelf. A one-piece vessel with an integral transparent lid does exist in food packaging — hinged sauce cups, for instance — but those are moulded in **polypropylene**, which is semi-crystalline and **hazy**. Haze is wide-angle scatter: precisely the pedestal chapter 5 is about, fed straight back into the beam. They are also tapered for stacking, with no flat reference surface, which is the opposite of what a repeatable optical seating needs. Disposable spectrophotometer cuvettes are made of polystyrene or PMMA exactly because those are optically clear.

Second — and this is decisive — the vessel sits above mains electricity. The lamp is a **220 V** unit housed in the **lower cone**, directly beneath the sample, because the beam runs top-down. **Any leak runs down into it.**

> ⇒ **Vessel integrity is a safety property of this instrument, not a measurement property.** A milled glass lid, a clamped polymer window, or any workshop adaptation is a *fabricated seal of unknown reliability sitting over mains voltage* — and the more aggressive the solvent it is asked to contain, the worse the consequence of a failure. **Any future vessel change must be argued on leak risk first and optics second.** The sealed, single-piece, known-good jar in use today is doing more work than it is usually given credit for.

This is also what closes the acetone option of §4.6: the chemistry is attractive, but it requires a new vessel, and every available new vessel is either optically worse or a hand-made seal over a 220 V lamp.

The change that would dissolve the constraint is a rebuild to **side illumination** — a horizontal beam, with the lamp no longer beneath the sample. That would also permit standard cuvettes, fixing path length and seating at the same time. It is a genuine option and a real redesign of the instrument; it is not currently planned.

7. What follows for the measurement

Six consequences, each traceable to a chapter above.

1. **Measure a ratio, never an absolute absorbance.** Concentration, path length and every multiplicative instrument property cancel (§2.3). This is the foundation everything else rests on.
2. **Keep the strongest band below about `A = 1`.** Beyond that, stray light flattens the response and the band stops reporting concentration honestly (§2.2).
3. **Treat sample clarity as an instrument parameter with its own error budget.** On our own measurements it is worth more than any optical improvement currently available (§5.4).
4. **Prefer clarifying to waiting.** Settling is slow, non-monotonic, and reversible with a knock (§4.4, §5.5). The analytical literature does not wait; it filters.
5. **Choose the solvent on two axes, not one.** Solvency for a triglyceride follows chain length; band shifts follow dielectric constant. They can be traded independently, and a good choice gets dissolution without moving the bands (§4.5).
6. **Remember that the vessel and the solvent are one decision.** Polystyrene and alcohols go together; hydrocarbons require replacing both (§6.1–6.2). And watch the vessel optically, not structurally — a crazing jar mimics a turbid sample (§6.3).

> **What is still open.** Whether a better solvent removes the pedestal in practice; how much of the remaining sample-to-sample variation is preparation rather than instrument; and whether a window reaching past 700 nm — where the strong chlorophyll band lies, and where a genuinely pigment-free region would finally exist — is worth the optical change. These are owned by `SPEC_capture_quality.md` §16.12 and `SPEC_capability_proof.md` §11.4.

Appendix — sources and further reading

Absorption spectroscopy and Beer–Lambert. Any physical-chemistry text; the treatment here follows the standard one. Porphyrin band structure: M. Gouterman, *Spectra of porphyrins*, J. Mol. Spectrosc. **6** (1961) 138 — the four-orbital model that names the Soret and Q bands.

Chlorophyll and its degradation. Pheophytinisation and the colour of cooked greens are covered in any food-chemistry text. Pumpkin-specific composition: Fruhwirth & Hermetter (2007), *Seeds and oil of the Styrian oil pumpkin*, Eur. J. Lipid Sci. Technol. **109**(11) 1128–1140, 10.1002/ejlt.200700105.

Dichromatism. S. Kreft & M. Kreft, on the dichromaticity index and pumpkin seed oil as its type example.

Oil–alcohol miscibility. Rao & Arnold (1957), *Alcoholic extraction of vegetable oils IV — solubilities of vegetable oils in aqueous 2-propanol*, JAOCS, 10.1007/BF02637892. The critical-solution-temperature data behind §4.2.

The ouzo effect. Vitale & Katz (2003) for the original characterisation; for a modern direct observation of the droplet nucleation, *ACS Central Science* (2023), 10.1021/acscentsci.2c01194.

Scattering. Rayleigh and Mie regimes: any optics text. For the practical spectroscopic consequence, see the turbidity-correction literature on baseline fitting.

Analytical methods for oil pigments. A spectrophotometric method for plant pigments, Chem. Papers; and the standard cyclohexane-based determinations of olive-oil chlorophylls and carotenoids.

Materials and solubility. C. M. Hansen, *Hansen Solubility Parameters — A User's Handbook* (2nd ed., CRC Press 2007) — the three-parameter model, the polymer-sphere construction and the RED metric of §6.2, together with the tabulated values used there. Polystyrene and PMMA chemical-compatibility charts are also in circulation but disagree on marginal cases; prefer the parameter calculation, and soak-test rather than trust either. FEP optical data, AdTech transmission tables.

Our own measurements, wherever marked *(measured)*: `SPEC_capability_proof.md` §11 and `SPEC_capture_quality.md` §16, with the underlying reasoning in `KB_spectroscopy_physics.md`. The instrument half of the story is in the companion document, *Capture Fidelity*.
